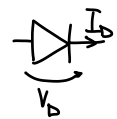


Formulario esteso: Analogica

DIODO



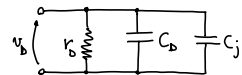
Modello esponenziale

$$I_D = I_S \left(e^{\frac{V_D}{V_{Th}}} - 1 \right)$$

Modello a soglia

$$\begin{cases} I_D = \phi & \text{per } V < V_\gamma \\ V_D = V_\gamma & \text{per } I_D > \phi \end{cases}$$

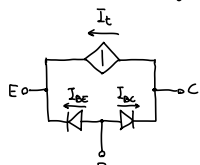
Modello piccoli segnali:



$$r_D \approx \begin{cases} \frac{I_{DQ}}{V_{Th}} & \text{in diretta} \\ \phi & \text{in inversa} \end{cases}$$

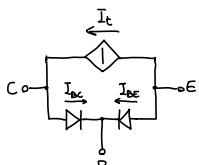
$$C_D = \tau_T Q_D \quad C_j = C_{j0} \frac{1}{\sqrt{1 - \frac{V_{DQ}}{\phi_j}}}$$

BJT



NPN

PNP



$$I_E = I_S \left[e^{\left(\frac{V_{BE}}{V_{Th}} \right)} - e^{\left(\frac{V_{BC}}{V_{Th}} \right)} \right]$$

$$= \beta_F I_{BE} - \beta_R I_{BC}$$

In reg. normale: $I_C = I_S e^{\frac{V_{BE}}{V_{Th}}}$

con Early: $I_C = I_S e^{\frac{V_{BE}}{V_{Th}}} \left(1 + \frac{V_{CE}}{V_A} \right) \sim V_{CE}$

$$= I_D \beta_{FO} \left(1 + \frac{V_{CE}}{V_A} \right) \sim V_{CE}$$

$$\beta_F$$

$$I_E = I_S \left[e^{\left(\frac{V_{EB}}{V_{Th}} \right)} - e^{\left(\frac{V_{CB}}{V_{Th}} \right)} \right]$$

$$= \beta_F I_{EB} - \beta_R I_{CB}$$

In reg. normale: $I_C = I_S e^{\frac{V_{EB}}{V_{Th}}}$

con Early: $I_C = I_S e^{\frac{V_{EB}}{V_{Th}}} \left(1 + \frac{V_{CE}}{V_A} \right) \sim V_{CE}$

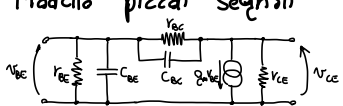
$$= I_D \beta_{FO} \left(1 + \frac{V_{CE}}{V_A} \right) \sim V_{CE}$$

$$\beta_F$$

MODELLO A SOGLIA

$$\begin{cases} V_{BE} < V_{BE,\gamma} \rightarrow \text{Interdizione: current: nulle} \\ V_{BE} > V_{BE,\gamma} \rightarrow \text{Saturazione: } V_{CE} = V_{CE,sat} \\ V_{BE} > V_{BE,\gamma} \rightarrow \text{Normale: } I_C = \beta_F I_B \end{cases}$$

Modello piccoli segnali



$$r_{BE} = \frac{V_{Th}}{I_{BEQ} + I_{BES}}$$

$$r_{BC} = \frac{V_{Th}}{I_{BCQ} + I_{BES}}$$

regione normale $\rightarrow q_{m,r_{BE}} = \beta_F \approx \beta_F$

$$q_{m,r_{BE}} = \beta_F i_B$$

MODELLO A SOGLIA

$$\begin{cases} V_{EB} < V_{EB,\gamma} \rightarrow \text{Interdizione: current: nulle} \\ V_{EB} > V_{EB,\gamma} \rightarrow \text{Saturazione: } V_{CE} = V_{CE,sat} \\ V_{EB} > V_{EB,\gamma} \rightarrow \text{Normale: } I_C = \beta_F I_B \end{cases}$$

Modello piccoli segnali

$$r_{BE} = \frac{V_{Th}}{I_{BEQ} + I_{BES}} \quad C_{BE} = C_D + C_j \quad q_{m,r_{BE}} \approx \beta_F \frac{1}{V_{BE}} - \beta_{FO} \frac{I_{BES}}{V_A} - \beta_R \frac{1}{V_{BC}}$$

$$r_{BC} = \frac{V_{Th}}{I_{BCQ} + I_{BES}} \quad C_{BC} = C_D + C_j \quad q_{m,r_{BC}} \approx \beta_{FO} \frac{I_{BES}}{V_A} + \beta_R \frac{1}{V_{BC}}$$

$$q_{m,r_{BE}} = \beta_F \approx \beta_F \quad q_{m,r_{BC}} = \frac{I_{CQ}}{V_{Th}}$$

MOSFET

n-MOSFET

p-MOSFET

$$V_{SB} > 0:$$

$$V_T = V_{T0} + \gamma \left(\sqrt{2\psi_F + V_{SB}} + \sqrt{2\psi_F} \right)$$

con $\psi_S = 2V_{Th} \ln \frac{N_A}{N_i} = 2\psi_F + V_{SB}$
potenziale superficiale

$$V_{GS} < V_T \rightarrow \text{MOSFET OFF}$$

$$V_{GS} > V_T \rightarrow \text{MOSFET in inversione}$$

$$V_{DS} \leq V_{GS} - V_T \quad \text{TRIODO}$$

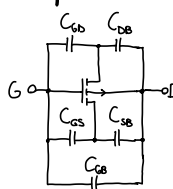
$$I_{DS} = \beta_n \left[(V_{GS} - V_T) V_{DS} - \frac{1}{2} V_{DS}^2 \right] (1 + \alpha V_{DS})$$

modulazione lunghezza di canale

$$V_{DS} > V_{GS} - V_T \quad \text{SATURAZIONE}$$

$$I_{DS} = \frac{\beta_n}{2} (V_{GS} - V_T)^2 (1 + \alpha V_{DS})$$

Capacitor nel MOSFET



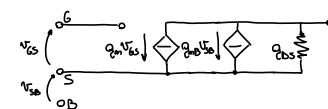
Reg. funz.	C_{GS}	C_{GD}	C_{DS}
OFF	ϕ	ϕ	WLC_{ox}
TRIODO	$\frac{1}{2} WLC_{ox}$	$\frac{1}{2} WLC_{ox}$	ϕ
SATUR.	$\frac{2}{3} WLC_{ox}$	ϕ	ϕ

$$C_{SD} = C_{DS} = W L_S C_j$$

con $C_j = C_{j0} \frac{1}{\sqrt{1 + \frac{|V_{SD}|}{\phi_j}}}$

$$C_{GSF} = C_{GDF} = W L C_{GS0}$$

Modello a piccolo segnali

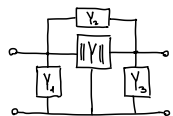


$$q_{mB} = -q_{m\alpha} \quad \alpha = \frac{\gamma}{2\sqrt{2\psi_F + V_{SBO}}}$$

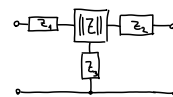
TRIODO: $q_m = \beta_n (1 + \alpha V_{DS0}) V_{DS0} \quad q_{DS} = \beta_n (V_{GS0} - V_T - V_{DS0}) (1 + \alpha V_{DS0}) + \frac{\alpha I_{DS0}}{1 + \alpha V_{DS0}}$

SATURAZIONE: $q_m = \beta_n (V_{GS0} - V_T) (1 + \alpha V_{DS0}) \quad q_{DS} = \alpha \frac{\beta_n}{2} (V_{GS0} - V_T)^2 = \frac{\alpha I_{DS0}}{1 + \alpha V_{DS0}}$

Matrici Linearizzazione



$$\|Y'\| = \begin{vmatrix} y_i + Y_1 + Y_3 & y_r - Y_3 \\ y_f - Y_3 & y_o + Y_2 + Y_3 \end{vmatrix}$$



$$\|Z'\| = \begin{vmatrix} z_i + z_1 + z_3 & z_r + z_3 \\ z_f + z_3 & z_o + z_2 + z_3 \end{vmatrix}$$

	z	y	h
A_r	$\frac{z_f z_c}{D_z + z_i z_c}$	$-\frac{y_f}{y_o + Y_c}$	$-\frac{h_f}{D_h + h_i Y_c}$
A_i	$-\frac{z_f}{z_o + z_c}$	$\frac{y_f Y_c}{D_y + y_i Y_c}$	$\frac{h_f}{1 + h_o z_c}$
$Z_i = \frac{1}{Y_i}$	$Z_i = z_i - \frac{z_r z_f}{z_o + z_c}$	$Y_i = y_i - \frac{y_r y_f}{y_o + Y_c}$	$Z_i = h_i - \frac{h_r h_f}{h_o + Y_c}$
$Z_o = \frac{1}{Y_o}$	$Z_o = z_o - \frac{z_r z_f}{z_i + z_2}$	$Y_o = y_o - \frac{y_r y_f}{y_i + Y_2}$	$Y_o = h_o - \frac{h_r h_f}{h_i + z_2}$

$$Z: \begin{cases} V_1 = Z_{11} I_1 + Z_{12} I_2 \\ V_2 = Z_{21} I_1 + Z_{22} I_2 \end{cases} \quad \begin{vmatrix} z_i & z_r \\ z_f & z_o \end{vmatrix}$$

$$G: \begin{cases} I_1 = Y_{11} V_1 + Y_{12} V_2 \\ I_2 = Y_{21} V_1 + Y_{22} V_2 \end{cases} \quad \begin{vmatrix} y_i & y_r \\ y_f & y_o \end{vmatrix}$$

$$h: \begin{cases} V_1 = h_{11} I_1 + h_{12} V_2 \\ I_2 = h_{21} I_1 + h_{22} V_2 \end{cases} \quad \begin{vmatrix} h_i & h_r \\ h_f & h_o \end{vmatrix}$$

Calcolo poli: $1 + SX$

$$f_p = \frac{1}{2\pi X}$$

n-MOSFET:

p-MOSFET:

Formulario esteso: Digitale

Potenza statica $\rightarrow P = V_{CC} I_{CC}$

Potenza dinamica $\rightarrow P_{din} = V_{CC} C_L \frac{V_{OH} - V_{OL}}{V_{SWING}} f_{0 \rightarrow 1}$

Inverter CMOS

$$V_{LT} = \frac{V_{TN} + \sqrt{\frac{\beta_p}{\beta_n} (V_{DD} + V_{TP})}}{1 + \sqrt{\frac{\beta_p}{\beta_n}}} \quad \left(\begin{array}{l} \text{vogliamo } V_{LT} = V_{DD}/2 \\ \text{ho p-Mosfet} \\ \text{grandi il doppio} \end{array} \right) \quad t_r = \frac{2 C_L}{\beta_p V_{DD}} F\left(\frac{|V_{TP}|}{V_{DD}}\right) \quad t_f = \frac{2 C_L}{\beta_n V_{DD}} F\left(\frac{V_{TN}}{V_{DD}}\right)$$

$$F(r) = \frac{1}{1-r} \left[\frac{r}{1-r} + \frac{1}{2} \ln(19-20r) \right] \quad t_p = \frac{C_L}{V_{DD}} \left(\frac{1}{\beta_n} + \frac{1}{\beta_p} \right) F\left(\frac{V_T}{V_{DD}}\right)$$

$$P_{statica} = \phi \quad P_{dinamica} = C_L V_{DD}^2 f_{0 \rightarrow 1}$$

$$C_{Gn} = W_n L_n C_{ox} + 2 W_n C_{GSO} \quad C_{Gp} = W_p L_p C_{ox} + 2 W_p C_{GSO}$$

$$C_{inv} = (W_n + W_p) L_{min} C_{ox} + 2(W_n + W_p) C_{GSO} \quad S_n = \frac{W_n}{L_{min}} \quad S_p = \frac{W_p}{L_{min}} \quad \alpha = \frac{S_p}{S_n}$$

$$= (S_n + S_p) C_{M1}$$

$$C_{M1} = L_{min}^2 C_{ox} + 2 L_{min} C_{GSO}$$

Gate statici CMOS

$$PU(\bar{I}_1, \dots, \bar{I}_n) = F(I_1, \dots, I_n)$$

$$PD(I_1, \dots, I_n) = \overline{F(I_1, \dots, I_n)}$$

Serie di MOSFET $\rightarrow S_1 // S_2$

Parallelo di MOSFET $\rightarrow S_1 + S_2$

$$\text{Self-loading} \rightarrow C_L = \overbrace{k C_{in, valle}}^{\text{capacitor a valle}} + C_W + S_n \underbrace{(n + m \alpha)}_{\substack{\text{parte n} \\ \text{all'ingresso}}} C_{p1}$$

uscite

parte p all'ingresso

$$C_{p1} = L_{min} \left[2 C_{GSO} + \frac{1}{2} L_{min} C_{ox} R_{DD} + K_{eq} L_S C_{j0} \right]$$

$$R_{DD} = 1 - \frac{V_T}{V_{DD}} \quad K_{eq} = \frac{2 \Phi_j}{V_{DD}} \left[\sqrt{1 + \frac{V_{DD}}{\Phi_j}} - 1 \right]$$

$$f_{0 \rightarrow 1} = f_{CLK} P_{0 \rightarrow 1} \quad P_{0 \rightarrow 1} = P_0 \cdot P_1$$

con prob. degli ingressi 0 e 1 uguali $\rightarrow$

$$P_0 = \frac{\text{out} = 0}{\text{num. out}}$$

$$P_1 = \frac{\text{out} = 1}{\text{num. out}}$$

se sono diverse $\rightarrow$

$$P_0 = P_A P_0$$

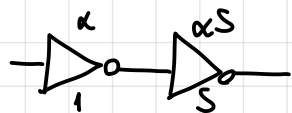
$$P_1 = 1 - P_0$$

Pilotaqgio grandi capacità

$$t_{po} = \frac{2(1+\alpha)C_{M1}}{\beta_n' V_{DD}} F\left(\frac{V_T}{V_{DD}}\right)$$

$$t_p = \frac{C_L}{C_{inv}} t_{po}$$

$$X = \frac{C_L}{C_{inv}}$$



$$t_p = t_{po} \cdot \left(5 + \frac{X}{S}\right)$$

$$S_{opt} = \sqrt{X}$$

conviene se $X > 4$

N buffer

$$N_{opt} = \frac{\ln X}{\ln U}$$

dove U = rapporto costante tra C uscita ed ingresso

$$U_{opt} = e$$

Pass-transistor

complementary (p ed n in parallelo) $\rightarrow R_{eq} = \frac{2V_{DD}}{\beta_n(V_{DD}-V_{Tno})^2 + \beta_p(V_{DD}-|V_{Tpo}|)^2}$

tempo carica 90% $t_{r,x} = \ln 10 R_{eq} C_x$ $V_x = V_{DD} \left[1 - e^{\left(-\frac{t}{R_{eq} C_x}\right)} \right]$

per n pass-transistor identici in serie $\rightarrow T_n = R_{eq} C \frac{n(n-1)}{2}$

serie di n trans.

da spezzare in blocchi $\rightarrow n/m$ blocchi da m transistor + $(n/m - 1)$ buffer da m trans.

$$m_{opt.} = \sqrt{\frac{2T_{buffer}}{2,3 \cdot R_{eq} C}} \approx 3$$

Logiche dinamiche CMOS

$P_{din} \rightarrow$ se precarico $P_1 = 1$, se pre-scarico $P_0 = \phi$

Extra

$$R_o \text{ inverter} \rightarrow g_o = S \beta' (V_{DD} - V_T) = \frac{1}{R_o}$$

n-MOSFET

$$V_{DS} \leq V_{GS} - V_T \quad \text{TRIODO}$$

$$I_{DS} = \beta_n \left[(V_{GS} - V_T) V_{DS} - \frac{1}{2} V_{DS}^2 \right] (1 + \lambda V_{DS})$$

modulazione
lunghezza di canale

$$V_{DS} > V_{GS} - V_T \quad \text{SATURAZIONE}$$

$$I_{DS} = \frac{\beta_n}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$